

Secondary aerosolization of viable *Bacillus anthracis* spores in a contaminated US Senate Office.

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Evidence abstract

CONTEXT Bioterrorist attacks involving letters and mail-handling systems in Washington, DC, resulted in *Bacillus anthracis* (anthrax) spore contamination in the Hart Senate Office Building and other facilities in the US Capitol's vicinity.

OBJECTIVE To provide information about the nature and extent of indoor secondary aerosolization of *B anthracis* spores.

DESIGN Stationary and personal air samples, surface dust, and swab samples were collected under semiquiescent (minimal activities) and then simulated active office conditions to estimate secondary aerosolization of *B anthracis* spores.

Nominal size characteristics, airborne concentrations, and surface contamination of *B anthracis* particles (colony-forming units) were evaluated.

RESULTS Viable *B anthracis* spores reaerosolized under semiquiescent conditions, with a marked increase in reaerosolization during simulated active office conditions.

Increases were observed for *B anthracis* collected on open sheep blood agar plates ($P < .001$) and personal air monitors ($P = .01$) during active office conditions.

More than 80% of the *B anthracis* particles collected on stationary monitors were within an alveolar respirable size range of 0.95 to 3.5 μm .

CONCLUSIONS *Bacillus anthracis* spores used in a recent terrorist incident reaerosolized under common office activities.

These findings have important implications for appropriate respiratory protection, remediation, and reoccupancy of contaminated office environments.

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